

Aaron Gember-Jacobson

Associate Professor & Chair
Department of Computer Science, Colgate University
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EDUCATION

Ph.D. in Computer Science <i>University of Wisconsin–Madison, Madison, WI</i>	May 2016
Master of Science in Computer Science <i>University of Wisconsin–Madison, Madison, WI</i>	May 2011
Bachelor of Science in Computer Science <i>Marquette University, Milwaukee, WI</i>	May 2009

EMPLOYMENT

Associate Professor , <i>Colgate University</i>	July 2023 – Present
Computer Science Department Chair , <i>Colgate University</i>	July 2023 – June 2026
Assistant Professor , <i>Colgate University</i>	July 2016 – June 2023
Lecturer , <i>University of Wisconsin–Madison</i>	Spring 2015, Spring 2014
Research Assistant/Fellow , <i>University of Wisconsin–Madison</i>	January 2010 – May 2016
Student Intern , <i>AT&T Research</i>	May 2011 – July 2011

TEACHING

([†] indicates course includes a weekly 2 hour laboratory in addition to lecture)

COLGATE UNIVERSITY — *undergraduate liberal arts institution with a 5 course teaching load*

- **Intro to Computing I (COSC 101)**: Fall 2024, Fall 2021, Spring 2020[†], Spring 2018[†]
- **Intro to Computing II (COSC 102)**: Spring 2023[†]
- **Intro to Computer Systems (COSC 208)**: Fall 2025, Fall 2024, Spring 2024, Fall 2023 (2 sections), Spring 2023[†], Fall 2022 (2 sections)[†], Spring 2022[†], Fall 2021[†], Spring 2021 (2 sections)[†], Fall 2020[†]
- **Operating Systems (COSC 301)**[†]: Spring 2025, Spring 2019, Fall 2018, Fall 2017, Spring 2017, Fall 2016 (2 sections)
- **Computer Networks (COSC 465)**: Spring 2026, Spring 2024, Spring 2022[†], Spring 2020[†], Spring 2019[†], Spring 2017
- **The Unreliable Internet (CORE S190)**: Spring 2026, Fall 2020 (FSEM), Fall 2018 (FSEM)

UNIVERSITY OF WISCONSIN-MADISON — *courses taught as a lecturer during PhD program*

- **Introduction to Networks (CS 640)**: Spring 2015, Spring 2014

PUBLISHED WORKS

(* indicates Colgate student author)

REFEREED JOURNAL PUBLICATIONS

- J2. Xi Jiang, Aaron Gember-Jacobson, and Nick Feamster. CAP: Detecting network device misconfigurations with context-aware prompting of LLMs. *Proc. ACM Meas. Anal. Comput. Syst.*, 10(2), 2026
- J1. Xi Jiang, Shinan Liu, Aaron Gember-Jacobson, Arjun Nitin Bhagoji, Paul Schmitt, Francesco Bronzino, and Nick Feamster. NetDiffusion: Network data augmentation through protocol-constrained traffic generations. *Proceedings of the ACM on Measurement and Analysis of Computing Systems (POMACS)*, 8(1):20, February 2024

REFEREED CONFERENCE PUBLICATIONS

- C18. Dan Wang, Peng Zhang, and Aaron Gember-Jacobson. Espresso: Control plane verification considering external routes. In *Proceedings of the ACM SIGCOMM 2024 Conference (SIGCOMM)*, pages 197–212. ACM, 2024 🏆 **Best Paper Honorable Mention**
- C17. Peng Zhang, Dan Wang, and Aaron Gember-Jacobson. Symbolic router execution. In *Proceedings of the ACM SIGCOMM 2022 Conference (SIGCOMM)*, pages 336–349. ACM, 2022
- C16. Peng Zhang, Aaron Gember-Jacobson, Yueshang Zuo, Yuhao Huang, Xu Liu, and Hao Li. Differential network analysis. In *Proceedings of the 19th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, pages 601–615. USENIX, 2022
- C15. Anubhavnidhi Abhashkumar, Aaron Gember-Jacobson, and Aditya Akella. AED: Incrementally synthesizing policy-compliant and manageable configurations. In *Proceedings of the 16th International Conference on Emerging Networking EXperiments and Technologies (CoNEXT)*, pages 482–495. ACM, 2020
- C14. Anubhavnidhi Abhashkumar, Aaron Gember-Jacobson, and Aditya Akella. Tiramisu: Fast multilayer network verification. In *Proceedings of the 17th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, pages 201–219. USENIX Association, 2020
- C13. Aaron Gember-Jacobson, Aditya Akella, Ratul Mahajan, and Hongqiang Harry Liu. Automatically repairing network control planes using an abstract representation. In *Proceedings of the 26th Symposium on Operating Systems Principles (SOSP)*, pages 359–373. ACM, 2017
- C12. Aaron Gember-Jacobson, Raajay Viswanathan, Aditya Akella, and Ratul Mahajan. Fast control plane analysis using an abstract representation. In *Proceedings of the ACM SIGCOMM 2016 Conference (SIGCOMM)*, pages 300–313. ACM, 2016
- C11. Junaid Khalid, Aaron Gember-Jacobson, Roney Michael, Anubhavnidhi Abhashkumar, and Aditya Akella. Paving the way for NFV: Simplifying middlebox modifications using StateAlyzr. In *Proceedings of the 13th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, pages 239–253. USENIX Association, 2016
- C10. Aaron Gember-Jacobson, Wenfei Wu, Xiujun Li, Aditya Akella, and Ratul Mahajan. Management plane analytics. In *Proceedings of the 2015 ACM Internet Measurement Conference (IMC)*, pages 395–408. ACM, 2015

- C9. Keqiang He, Junaid Khalid, [Aaron Gember-Jacobson](#), Sourav Das, Chaithan Prakash, Aditya Akella, Li Erran Li, and Marina Thottan. Measuring control plane latency in SDN-enabled switches. In *Proceedings of the 1st ACM SIGCOMM Symposium on Software Defined Networking Research (SOSR)*, pages 25:1–25:6. ACM, 2015
- C8. Ashok Anand, [Aaron Gember-Jacobson](#), Collin Engstrom, and Aditya Akella. Design patterns for tunable and efficient SSD-based indexes. In *Proceedings of the tenth ACM/IEEE symposium on Architectures for networking and communications systems (ANCS)*, pages 149–160. ACM, 2014
- C7. [Aaron Gember-Jacobson](#), Raajay Viswanathan, Chaithan Prakash, Robert Grandl, Junaid Khalid, Sourav Das, and Aditya Akella. OpenNF: enabling innovation in network function control. In *Proceedings of the ACM SIGCOMM 2014 Conference (SIGCOMM)*, pages 163–174. ACM, 2014
- C6. Thomas Ball, Nikolaj Bjørner, [Aaron Gember](#), Shachar Itzhaky, Aleksandr Karbyshev, Mooly Sagiv, Michael Schapira, and Asaf Valadarsky. VeriCon: towards verifying controller programs in software-defined networks. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pages 282–293. ACM, 2014
- C5. Keqiang He, Alexis Fisher, Liang Wang, [Aaron Gember](#), Aditya Akella, and Thomas Ristenpart. Next stop, the cloud: understanding modern web service deployment in EC2 and Azure. In *Proceedings of the 2013 Internet Measurement Conference (IMC)*, pages 177–190. ACM, 2013
- C4. [Aaron Gember](#), Aditya Akella, Jeffrey Pang, Alexander Varshavsky, and Ramón Cáceres. Obtaining in-context measurements of cellular network performance. In *Proceedings of the 12th ACM SIGCOMM Internet Measurement Conference (IMC)*, pages 287–300. ACM, 2012
- C3. [Aaron Gember](#), Christopher Dragga, and Aditya Akella. ECOS: leveraging software-defined networks to support mobile application offloading. In *Proceedings of the Symposium on Architecture for Networking and Communications Systems (ANCS)*, pages 199–210. ACM, 2012
- C2. Shan-Hsiang Shen, [Aaron Gember](#), Ashok Anand, and Aditya Akella. Refactor-ing content overhearing to improve wireless performance. In *Proceedings of the 17th Annual International Conference on Mobile Computing and Networking (MobiCom)*, pages 217–228. ACM, 2011
- C1. [Aaron Gember](#), Ashok Anand, and Aditya Akella. A comparative study of handheld and non-handheld traffic in campus Wi-Fi networks. In *Proceedings of the 12th International Passive and Active Measurement Conference (PAM)*, pages 173–183. Springer, 2011

REFEREED WORKSHOP PUBLICATIONS

- W10. Xi Jiang, Shinan Liu, [Aaron Gember-Jacobson](#), Paul Schmitt, Francesco Bronzino, and Nick Feamster. Generative, high-fidelity network traces. In *Proceedings of the 22nd ACM Workshop on Hot Topics in Networks (HotNets)*, page 6. ACM, 2023
- W9. Emily Huff* and [Aaron Gember-Jacobson](#). Divesting in socially (ir)responsible internet service providers. In *Proceedings of the ACM SIGCOMM 2021 Workshop on Technologies, Applications, and Uses of a Responsible Internet (TAURIN)*, pages 21–28. ACM, 2021
- W8. Peng Zhang, Yhao Huang, [Aaron Gember-Jacobson](#), Wenbo Shi, Xu Liu, Hongkun Yang, and Zhiqiang Zuo. Incremental network configuration verification. In *Proceedings of the 19th ACM Workshop on Hot Topics in Networks (HotNets)*, pages 81–87. ACM, 2020
- W7. [Aaron Gember-Jacobson](#), Costin Raiciu, and Laurent Vanbever. Integrating verification and repair into the control plane. In *Proceedings of the 16th ACM Workshop on Hot Topics in Networks (HotNets)*, pages 129–135. ACM, 2017

- W6. Junaid Khalid, Mark Coatsworth, Aaron Gember-Jacobson, and Aditya Akella. A standardized southbound API for VNF management. In *Proceedings of the ACM SIGCOMM Workshop on Hot topics in Middleboxes and Network Function Virtualization (HotMiddlebox)*, pages 38–43. ACM, 2016
- W5. Aaron Gember-Jacobson and Aditya Akella. Improving the safety, scalability, and efficiency of network function state transfers. In *Proceedings of the 2015 ACM SIGCOMM Workshop on Hot Topics in Middleboxes and Network Function Virtualization (HotMiddlebox)*, pages 43–48. ACM, 2015
- W4. Anand Krishnamurthy, Shoban P. Chandrabose, and Aaron Gember-Jacobson. Pratyaaastha: an efficient elastic distributed SDN control plane. In *Proceedings of the 3rd workshop on Hot topics in software defined networking (HotSDN)*, pages 133–138. ACM, 2014
- W3. Aaron Gember, Prathmesh Prabhu, Zainab Ghadiyali, and Aditya Akella. Toward software-defined middlebox networking. In *Proceedings of the 11th ACM Workshop on Hot Topics in Networks (HotNets)*, pages 7–12. ACM, 2012
- W2. Aaron Gember, Chris Dragga, and Aditya Akella. ECOS: practical mobile application offloading for enterprises. In *Proceedings of the 2nd USENIX Workshop on Hot Topics in Management of Internet, Cloud, and Enterprise Networks and Services (Hot-ICE)*. USENIX Association, 2012
- W1. Ashok Anand, Aaron Gember, Aditya Akella, and Vyas Sekar. Tracking semantic relationships for effective data management in home networks. In *Proceedings of the 2010 ACM SIGCOMM Workshop on Home Networks (HomeNets)*, pages 49–54. ACM, 2010

REFEREED POSTERS (EXTENDED ABSTRACTS)

([†] indicates conference did not include the extended abstracts in the published proceedings)

- P11. Brian Douglas Jr.* and Aaron Gember-Jacobson. Light security: Discovering vulnerabilities in theatrical lighting control systems and assessing their prevalence. In *21st Symposium on Usable Privacy and Security (SOUPS)*. USENIX, 2025
- P10. Danny Orlando* and Aaron Gember-Jacobson. Evaluating browser-based networking for real-time multiplayer games. In *22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI)*. USENIX, 2025[†]
- P9. Jyotirmay Chauhan* and Aaron Gember-Jacobson. Summarizing network configuration patterns. In *20th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*. USENIX, 2023[†] 🏆 **First place in the Undergraduate Student Research Competition**
- P8. Sara Alam* and Aaron Gember-Jacobson. Poster: Identifying syntactic motifs and errors in router configurations using graphs. In *30th IEEE International Conference on Network Protocols (ICNP)*. IEEE, 2022
- P7. Omshi Samal* and Aaron Gember-Jacobson. Testing router command line interfaces. In *ACM SIGCOMM 2022 Networking Networking Women Professional Development Workshop (N2Women)*. ACM, 2022[†]
- P6. Xi Jiang* and Aaron Gember-Jacobson. Non-interoperability detection for routing protocol implementations. In *ACM SIGCOMM 2021 Conference (SIGCOMM), Poster and Demo Sessions*, pages 4–5. ACM, 2021. (Awarded 1st place in the Undergraduate Student Research Competition)
- P5. Ruchit Shrestha*, Xiaolin Sun*, and Aaron Gember-Jacobson. Localizing router configuration errors using unsatisfiable cores. In *16th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*. USENIX, 2019[†]

- P4. Ahsan Mahmood* and Aaron Gember-Jacobson. Auto-completion for network configurations. In *15th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*. USENIX, 2018[†]
- P3. Keqiang He, Junaid Khalid, Sourav Das, Aaron Gember-Jacobson, Chaithan Prakash, Aditya Akella, Li Erran Li, and Marina Thottan. Latency in software defined networks: Measurements and mitigation techniques. In *Proceedings of the 2015 ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS)*, pages 435–436. ACM, 2015
- P2. Aaron Gember, Robert Grandl, Junaid Khalid, and Aditya Akella. Design and implementation of a framework for software-defined middlebox networking. In *Proceedings of the ACM SIGCOMM 2013 Conference (SIGCOMM)*, pages 467–468. ACM, 2013
- P1. Aaron Gember and Dennis Brylow. Real-Time TCP for Embedded Devices. *ACM Student Research Competition Poster Session, 40th ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2009.

TECHNICAL REPORTS (NON-REFEREED)

- R9. Xi Jiang, Aaron Gember-Jacobson, and Nick Feamster. CAIP: Detecting router misconfigurations with context-aware iterative prompting of LLMs. *arXiv Computing Research Repository (CoRR)*, abs/2411.14283, 2024. (12 pages)
- R8. Aaron Gember-Jacobson, Ruchit Shrestha*, and Xiaolin Sun*. Localizing router configuration errors using minimal correction sets. *arXiv Computing Research Repository (CoRR)*, abs/2204.10785, 2022. (15 pages)
- R7. Anubhavnidhi Abhashkumar, Aaron Gember-Jacobson, Aditya Akella, and Yibo Zhu. Tiramisu: Fast and general network verification. *arXiv Computing Research Repository (CoRR)*, abs/1906.02043, 2019. (17 pages)
- R6. Eric Campbell, Aaron Gember-Jacobson, Alex Horn, and Nate Foster. A two-stage network anomaly detection method using formal analysis. Technical Report 2482, Fujitsu Laboratories of America, 2018. (10 pages)
- R5. Aaron Gember, Robert Grandl, Junaid Khalid, Shan-Hsiang Shen, and Aditya Akella. Design and implementation of a framework for software-defined middlebox networking. Technical Report TR1794, University of Wisconsin-Madison, 2012. (14 pages)
- R4. Aaron Gember, Anand Krishnamurthy, Saul St. John, Robert Grandl, Xiaoyang Gao, Ashok Anand, Theophilus Benson, Aditya Akella, and Vyas Sekar. Stratos: A network-aware orchestration layer for middleboxes in the cloud. *arXiv Computing Research Repository (CoRR)*, abs/1305.0209, 2013. (14 pages)
- R3. Ashok Anand, Aaron Gember, and Aditya Akella. Generic design patterns for tunable and high-performance ssd-based index. Technical Report TR1778, University of Wisconsin-Madison, 2012. (13 pages)
- R2. Aaron Gember Robert Grandl, Ashok Anand, Theophilus Benson, and Aditya Akella. Stratos: Virtual middleboxes as first-class entities. Technical Report TR1771, University of Wisconsin-Madison, 2012. (15 pages)
- R1. Aaron Gember Ashok Anand and Aditya Akella. Handheld vs. non-handheld traffic: Implications for campus wifi networks. Technical Report TR1679, University of Wisconsin-Madison, 2010. (12 pages)

INVITED BLOG POSTS (NON-REFEREED)

- B2. Aaron Gember-Jacobson and Aditya Akella. The verification-synthesis spectrum. *Netverify*, August 2020. <https://netverify.fun/the-verification-synthesis-spectrum>
- B1. Aditya Akella and Aaron Gember-Jacobson. Talk to a network operator today! *Netverify*, July 2020. <https://netverify.fun/talk-to-a-network-operator-today>

PRESENTATIONS

CONFERENCE/WORKSHOP PRESENTATIONS

- Refereed Conference Publications: C17, C16, C13, C8, C7, C5, C4, C2, C1
- Refereed Workshop Publications: W9, W5, W3, W2

INVITED TALKS

- T7. Verifying the Effects of Network Configuration Changes. *Aria Operations for Networks Product Team Seminar, VMWare by Broadcom*, January 2024.
- T6. Change-Centric Network Configuration Analysis. *Network Operations and Internet Security (Noise) Lab, University of Chicago*, April 2023.
- T5. Detecting Configuration Errors Via Pattern Mining. *Network Verification Workshop (NetVerify)*, November 2021. (with Jyotirmay Chauhan*, Devon Lee*, and Emily Yu*)
- T4. The Verification-Synthesis Spectrum. *Colgate University Natural Sciences and Mathematics Colloquium Series 2020-2021*, April 2021.
- T3. Verifying and repairing network control planes using an abstract representation. *Cornell University Systems Lunch*, September 2019.
- T2. Localizing router configuration errors using correction sets. *ACM SIGCOMM 2019 Workshop on Networking and Programming Languages (NetPL)*, August 2019
- T1. Providing automatic insights into network requirements. *Hamilton College Computer Science Colloquium*, March 2019.

PANELS

- N1. Advances, Opportunities, and "Non-Traditional" Software-Defined Networking Control. *ACM SIGCOMM Symposium on SDN Research (SOSR) 2021*, October 2021.

GRANTS

EXTERNAL

- E2. *NeTS: Medium: Collaborative Research: Automatic Network Repair*. National Science Foundation (NSF), 2018-2022, \$170K (plus \$1,030K awarded to the University of Wisconsin-Madison).
Role: Principal Investigator
- E1. *AitF: Collaborative Research: Foundations of Intent-based Networking*. National Science Foundation (NSF), 2016-2019, \$60K (plus \$356K awarded to the University of Wisconsin-Madison).
Role: Principal Investigator

INTERNAL

- I11. *Illuminating the practices of Internet service providers*. Colgate University, Faculty-Initiated Summer Student Fellowship, Summer 2023, \$8400.
- I10. *Learning from network history*. Colgate University, Faculty-Initiated Summer Student Fellowship, Summer 2022, \$4200.
- I9. *Improving network management with machine learning*. Colgate University, Faculty-Initiated Summer Student Fellowship, Summer 2021, \$4725.
- I8. *Practical bug detection in software implementations of routing protocols*. Colgate University, Faculty-Initiated Summer Student Fellowship, Summer 2020, \$3800.
- I7. *Analyzing network configurations and inferring/verifying networking behavior*. Colgate University, Student Wage Grant, Fall Summer 2019, \$1882.
- I6. *Updating computer networks one step at a time*. Colgate University, Faculty-Initiated Summer Student Fellowship, Summer 2019, \$4050.
- I5. *Leveraging WiFi signals to identify smartphone proximity*. Colgate University, Student Wage Grant, Fall 2018, \$590.
- I4. *Quickly and clearly repairing computer networks*. Colgate University, Faculty-Initiated Summer Student Fellowship, Summer 2018, \$7200.
- I3. *Efficiently monitoring and analyzing events in a live network to accurately detect errors*. Colgate University, Student Wage Grant, Spring 2018, \$1672.
- I2. *Automatically fixing common causes of network problems*. Colgate University, Student Wage Grant, Summer 2017, \$2233.
- I1. *Improving auto-generated repairs for computer networks*. Colgate University, Faculty-Initiated Summer Student Fellowship, Summer 2017, \$9000.

HONORS AND AWARDS

- Internet Engineering Task Force (IETF) Applied Networking Research Prize (2015)
- Department of Computer Sciences Graduate Student Instructor Award (2014)
- IBM Ph.D. Fellowship (2013-2015)
- Wisconsin Alumni Research Foundation (WARF) Innovation Award Finalist (2012)
- National Science Foundation Graduate Research Fellowship Honorable Mention (2011)
- University of Wisconsin—Madison, Computer Science Summer Graduate Fellowship (2010)
- Third Place in the ACM Student Research Competition at SIGCSE 2009

PATENTS

- X4. Aditya Akella, Junaid Khalid, and [Aaron Gember-Jacobson](#). “State extraactor for middlebox management system.” US Patent # 11,895,177. Issued February 2, 2024.
- X3. [Aaron Gember-Jacobson](#), Chaithan Prakash, Raajay Viswanathan, Robert Grandl, Junaid Kahlid, Sourav Das, and Aditya Akella. “Cloud Architecture with State-Saving Middlebox Scaling.” US Patent # 9,705,785. Issued July 11, 2017.

- X2. Ashok Anand, Aaron Gember, and Aditya Akella. “High-Performance Indexing for Data-Intensive Systems.” US Patent # 9,612,955. Issued April 4, 2017.
- X1. Aaron Gember, Robert Grandl, Theophilus Benson, Ashok Anand, and Aditya Akella. “Cloud-Based Middlebox Management System.” US Patent # 9,104,492. Issued August 11, 2015.

SERVICE

RESEARCH COMMUNITY

- *Program committee member*, International Conference on emerging Networking EXperiments and Technologies (CoNEXT): 2026, 2025, 2024, 2021, 2018
- *Publicity co-chair*, ACM SIGCOMM Conference (SIGCOMM): 2026
- *Program committee member*, USENIX Symposium on Networked Systems Design and Implementation (NSDI): 2025
- *Steering committee chair*, ACM Symposium on SDN Research (SOSR): 2019–2022
- *General chair*, ACM Symposium on SDN Research (SOSR): 2022
- *Program co-chair*, ACM Symposium on SDN Research (SOSR): 2019
- *Program committee member*, ACM Symposium on SDN Research (SOSR): 2021, 2020
- *External reviewer*, International Conference on Computer Aided Verification (CAV): 2022
- *Artifact evaluation committee co-chair*, ACM SIGCOMM Conference (SIGCOMM): 2021
- *Reviewer*, IEEE/ACM Transactions on Networking (TON): 2021, 2016, 2014
- *Moderator*, SIGCOMM Slack Workspace: 2020
- *Panelist*, National Science Foundation (NSF): 2020, 2019, 2018
- *Poster committee member*, ACM SIGCOMM Conference (SIGCOMM): 2020, 2019
- *Program committee member*, ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems: 2018
- *Poster and Demo committee member*, ACM Symposium on SDN Research (SOSR): 2018
- *Program committee member*, Asia-Pacific Workshop on Networking (APNet): 2017
- *Program committee member*, IEEE International Symposium on Local and Metropolitan Area Networks (LANMAN): 2017
- *Reviewer*, ACM New York Celebration of Women in Computing (NYCWIC): 2017
- *External reviewer*, International Journal of Network Measurement (IJNM): 2016, 2014
- *External reviewer*, Symposium on Principles of Programming Languages (POPL): 2016
- *External reviewer*, Computer Communications Review (CCR): 2016, 2015
- *External reviewer*, IEEE Transactions on Wireless Communications (TWC): 2015
- *Poster committee member*, ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS): 2014

UNIVERSITY

- *Chair*, Nominating Committee: Fall 2024–Spring 2025
- *Elected committee member*, Nominating Committee: Fall 2023–Spring 2026

- *Elected committee member*, Faculty Affirmative Action Oversight Committee: Fall 2023–Spring 2024
- *Computer Science representative*, Bernstein Hall Users Group: Fall 2021–Spring 2025
- *Executive advisory committee member*, Picker Interdisciplinary Science Institute: Fall 2021–Spring 2024
- *Co-leader*, DEI in STEM: Spring 2023–Fall 2023
- *Group member*, DEI in STEM: Fall 2020–Fall 2024
- *Co-facilitator*, Dart Colgrove Commons Wellness Discussion Group: Fall 2020
- *Ex officio representative*, Committee on Information Technology (CIT): Fall 2018–Spring 19
- *Appointed member*, Fellowships Committee: Fall 2021, Fall 2018, Fall 2017
- *Search committe member*, Information Technology Services (ITS): Spring 2019, Spring 2017

DEPARTMENT

- *Department Chair*: Summer 2023–Summer 20226
- *Organizer*, Task Force on Diversity, Equity, and Inclusion (DEI) in Computer Science: Fall 2020–Spring 2022
- *Organizer*, weekly department symposium: Spring 2020, Spring 2017–Spring 2019
- *Search committee chair*: Fall 2024, Spring 2024, Spring 2023
- *Search committee member*: Spring 2023, Fall 2022, Spring 2022, Fall 2021, Spring 2021, Fall 2019, Spring 2019, Fall 2018, Spring 2018, Spring 2017, Fall 2017

RESEARCH MENTORING

BACHELOR'S THESES

- Danny Orlando '25: "Evaluating Browser-Based Networking for Real-Time Multiplayer Games"
- Tajveer Singh Dhesi '25: "Measuring the Variety of Website Arrival Age Gates in the E-Cigarette, Alcohol, and Gambling Industry"
- Omshi Samal '24: "Hunting for Bugs in Router Command Line Interfaces"
- Sara Alam '23: "Motif Mining"
- Zhaoyang Liu '23: "Historical Analysis"
- Jeffery Yang '23: "Making Pointers Fun"
- Xinai (Nikki) Zhang '23: "Comparing Unicode and Weibo Emoji Text Representations on Text Categorization Problems"
- Xi (Chase) Jiang '21: "Non-interoperability Detection for Routing Protocol Implementations"
- Emily Huff '20: "Socially Responsible Internet Routing"
- Xiaolin (Owen) Sun '20: "Energy-Aware Network Routing"
- Alex Thomas '19: "Examinations of Stock Trading Algorithms and Applications of Twitter Sentiment for Trading Strategies"
- Ahsan Mahmood '18: "Autocompletion for Network Configurations"

UNDERGRADUATE RESEARCH STUDENTS

- Ralph Hyacinthe '27 (Fall 2025–Present)

- Pule Motsamai '27 (Fall 2025-Present)
- Tatiwoine Sanders '26 (Fall 2025-Spring 2026)
- Brian Douglas Jr. '25 (Spring 2025)
- Tajveer Singh Dhesi '25 (Spring 2025)
- Aidan Fox '25 (Fall 2024-Spring 2025)
- Danny Orlando '25 (Fall 2024-Spring 2025)
- Omshi Samal '24 (Spring 2021, Fall 2021-Spring 2024)
- Jaanhvi Agarwal '25 (Summer 2023)
- Amanda Anowi '24 (Summer 2023)
- Sara Alam '23 (Spring 2022-Spring 2023)
- Jyotirmay Chauhan '23 (Summer 2020, Summer 2021-Spring 2023)
- Zhaoyang Liu '23 (Summer 2022-Spring 2023)
- Jeffery Yang '23 (Fall 2022-Spring 2023)
- Emily Yu '23 (Summer 2021-Fall 2021)
- Xinai (Nikki) Zhang '23 (Fall 2022-Spring 2023)
- Devon Lee '22 (Summer 2021-Spring 2022)
- Megan Imperato '21 (Spring 2019)
- Xi (Chase) Jiang '21 (Summer 2020-Spring 2021)
- Yasoob Khalid '21 (Summer 2018-Spring 2019, Fall 2020)
- Tyler Potter '21 (Summer 2020)
- Mary Festa '20 (Summer 2019-Fall 2019)
- Emily Huff '20 (Fall 2019-Spring 2020)
- Ruchit Shrestha '20 (Summer 2017-Summer 2019)
- Xiaolin (Owen) Sun '20 (Summer 2018-Summer 2020)
- Eliza Lucas '20 (Fall 2019-Spring 2020, Spring 2018-Fall 2018)
- Alex Thomas '19 (Spring 2018)
- Lindsey Derbyshire '18 (Summer 2017)
- Ahsan Mahmood '18 (Summer 2017-Spring 2018)
- Amanda Milberg '18 (Spring 2018)
- Franklin van Nes '18 (Summer 2017)
- Danielle Zegelstein '18 (Spring 2018)
- Francisco Flores '17 (Spring 2017)
- Alec Glassman '17 (Spring 2017)
- Saw Lin '17 (Spring 2017)
- Pam Needle '17 (Spring 2017)

PHD THESIS COMMITTEE

- Yongchao He (Tsinghua University, April 2023)
- Hongyi Huang (Tsinghua University, May 2022)
- Rüdiger Birkner (ETH Zurich, September 2021)